

Programming Test

Date: 16/04/2026

Instructions:

1. Solve any 4 questions.
2. Input should be from user.
3. Do not use any inbuilt functions.

Q1. Write a Java program to find the minimum number of jumps required to reach the last index.

Description

- Each element represents maximum jump length
- If not reachable, return -1

Input :- arr = [2, 3, 1, 1, 4]

Output :- 2

Q2. Write a Java program to find the maximum sum of a subarray where you are allowed to delete at most one element.

Description

- You can choose a continuous subarray.
- You may delete one element from the subarray to maximize the sum.
- If all elements are negative, return the maximum element.

Input :- arr = [1, -2, 0, 3]

Output :- 4

Q3. Write a Java program to partition a string such that every substring is a palindrome.

Description

- Return all possible palindrome partitions
- Use backtracking logic

Input :- str = "aab"

Output :- [["a", "a", "b"], ["aa", "b"]]

Q4. Write a Java program to check whether two strings are isomorphic.

Description

- Two strings are isomorphic if characters in one string can be replaced to get the second string
- Each character must map to only one character

Input :- s1 = "egg"

s2 = "add"

Output :- true

Q5. Write a Java program to generate all possible permutations of a given string.

Description

- A permutation is a rearrangement of characters
- All permutations must be unique
- Order of output does not matter

Input :- str = "ABC"

Output :- ABC ACB BAC BCA CAB CBA

-----ALL THE BEST-----